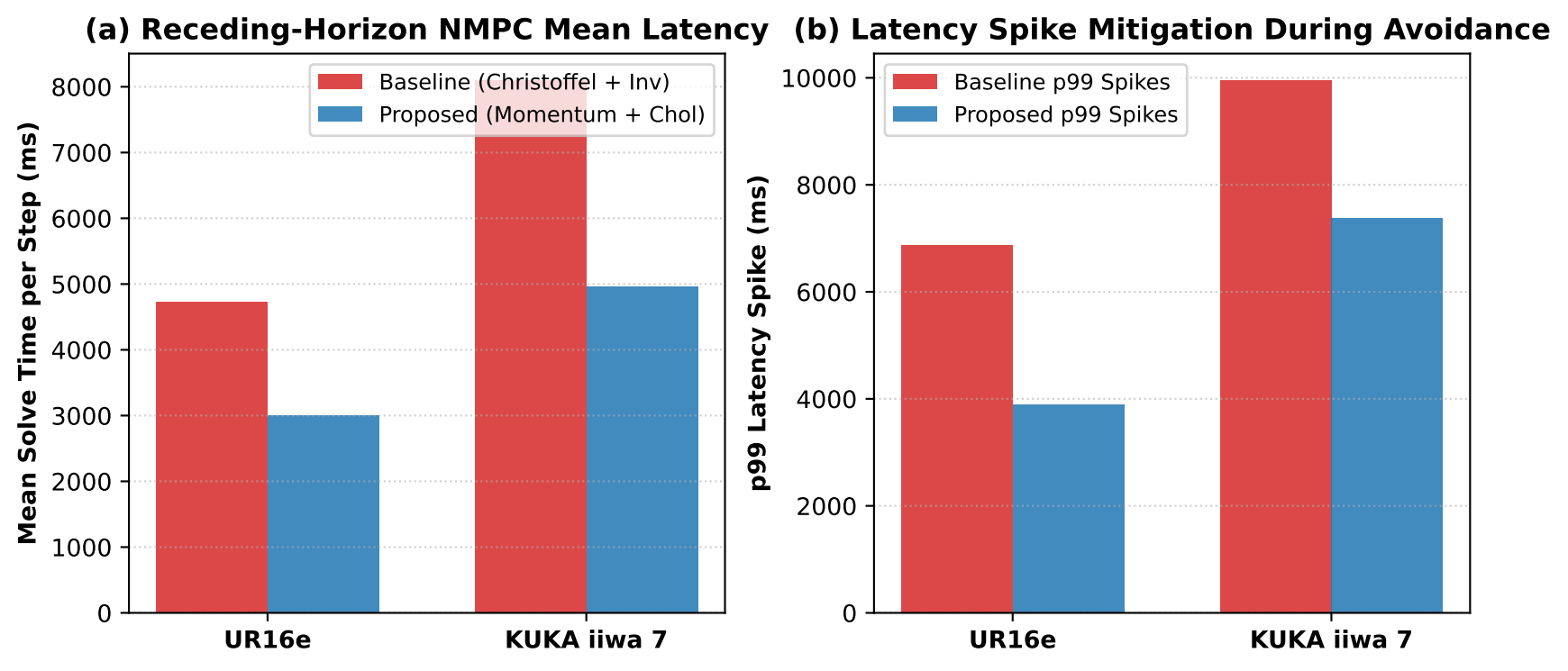
(a) Receding-Horizon NMPC Mean Latency
Mean Solve Time per Step (ms)
0
1000
2000
3000
4000
5000
6000
7000
8000
Baseline (Christoffel + Inv)
Proposed (Momentum + Chol)
UR16e
KUKA iiwa 7
(b) Latency Spike Mitigation During Avoidance
p99 Latency Spike (ms)
0
2000
4000
6000
8000
10000
Baseline p99 Spikes
Proposed p99 Spikes
UR16e
KUKA iiwa 7